

SUMBURGH, United Kingdom

WMO: 030030

Lat: 59.879N

Lon: 1.296W

Elev: 6

StdP: 101.25

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.2	-0.1	-6.0	2.3	0.8	-4.9	2.5	1.5	20.2	4.4	19.2	5.1	5.3	350	0.819

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.7	16.5	15.0	15.8	14.3	15.0	13.6	15.2	16.3	14.6	15.5	14.0	14.8	5.5	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.8	10.5	16.0	14.1	10.0	15.0	13.2	9.5	14.2	42.5	16.2	40.3	15.5	39.0	14.6	18.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.2	15.8	14.0	-3.3	18.0	1.7	1.4	-4.5	19.0	-5.5	19.8	-6.5	20.6	-7.7	21.6	
			DB	-4.2	16.3	1.4	1.1	-5.3	17.1	-6.1	17.8	-6.9	18.5	-8.0	19.3
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	5.2	4.8	5.2	6.6	8.3	10.7	12.7	12.9	11.7	9.8	7.1	5.9
	DBStd	3.53	2.15	2.21	2.23	1.95	1.84	1.39	1.52	1.28	1.78	2.02	2.55	2.69
	HDD10.0	863	151	146	149	103	57	8	1	0	7	27	88	127
	HDD18.3	3620	409	379	407	353	312	230	176	168	200	264	336	385
	CDD10.0	285	0	0	0	0	4	28	83	91	57	21	2	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.8	9.1	8.2	7.9	6.6	5.6	5.2	4.7	5.0	6.2	7.2	7.9	8.2
Precipitation	PrecAvg	1155	145	111	103	62	61	57	61	83	94	119	126	135
	PrecMax	1387	272	182	186	111	130	118	125	191	133	172	185	239
	PrecMin	944	47	45	28	28	15	13	22	37	33	55	54	60
	PrecStd	123	52	36	33	19	23	25	26	33	28	32	34	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.0	9.2	9.6	11.2	14.1	16.2	18.2	17.2	16.0	13.8	12.1	11.1
		MCWB	9.1	8.3	8.5	9.1	12.1	13.8	16.3	15.3	14.2	12.2	10.8	10.0
	2%	DB	9.1	9.0	9.1	10.2	12.8	14.1	16.9	16.2	15.1	13.1	11.2	10.2
		MCWB	8.4	8.3	8.2	8.6	10.8	12.4	15.6	14.6	13.6	11.8	10.1	9.4
	5%	DB	8.9	8.2	8.8	9.9	11.9	13.2	15.9	15.6	14.2	12.8	11.0	10.0
		MCWB	8.2	7.5	8.0	8.4	10.1	11.6	14.6	14.1	13.0	11.5	10.0	9.2
	10%	DB	8.1	7.9	8.1	9.1	11.0	12.9	15.0	15.0	14.0	12.1	10.2	9.2
		MCWB	7.3	7.2	7.2	8.0	9.3	11.3	13.7	13.6	12.8	11.0	9.2	8.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.2	8.9	8.7	9.9	12.7	14.6	16.6	15.7	14.7	12.8	11.4	10.4
		MCDB	9.2	9.0	8.9	10.4	13.8	16.3	17.9	16.7	15.7	13.3	11.8	10.9
	2%	WB	8.5	8.2	8.3	9.1	11.2	12.8	15.6	15.0	14.1	12.4	10.9	9.7
		MCDB	8.8	8.6	8.6	9.7	12.3	14.0	16.6	16.1	14.9	12.9	11.3	10.0
	5%	WB	8.0	7.7	7.8	8.6	10.4	12.2	14.8	14.5	13.5	11.9	10.2	9.2
		MCDB	8.4	8.2	8.2	9.2	11.4	13.3	15.7	15.4	14.4	12.6	10.7	9.6
	10%	WB	7.4	7.0	7.3	8.1	9.7	11.7	14.0	14.0	13.1	11.5	9.6	8.6
		MCDB	8.0	7.6	7.8	8.8	10.6	12.7	14.8	14.9	14.0	12.3	10.2	9.2
Mean Daily Temperature Range	5% DB	MDBR	3.1	3.0	3.3	3.7	4.2	3.7	3.6	3.7	3.3	3.1	3.2	3.2
		MCDBR	3.3	2.8	3.1	3.7	4.6	4.1	4.4	4.3	3.5	2.9	2.9	2.9
		MCWBR	3.7	3.0	3.0	2.8	3.3	2.9	3.1	3.2	2.8	2.8	3.5	3.5
	5% WB	MCDBR	3.1	2.7	2.8	3.2	4.4	3.8	4.1	3.9	3.2	2.7	2.9	2.9
MCWBR		3.6	3.1	3.0	2.8	3.3	2.8	3.2	3.1	2.7	3.0	3.6	3.7	
Clear-Sky Solar Irradiance	taub	0.338	0.356	0.370	0.387	0.375	0.371	0.385	0.371	0.366	0.351	0.334	0.429	
	taud	2.093	2.208	2.278	2.291	2.394	2.447	2.422	2.461	2.435	2.322	2.151	2.460	
	Ebn at Noon	394	622	750	810	852	862	838	823	757	627	399	274	
	Edh at Noon	49	74	93	109	106	102	102	92	80	65	45	34	
All-Sky Solar Radiation	RadAvg	0.25	0.80	1.94	3.52	4.92	4.98	4.55	3.71	2.27	1.13	0.37	0.16	
	RadStd	0.03	0.09	0.13	0.25	0.31	0.47	0.39	0.34	0.19	0.11	0.03	0.01	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (19 neighbors)	+0.32	N/A	+0.78	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page